

Applied Bioinformatics Laboratory

Experiment – 5: Crispr Efficiency Prediction Using Crisprpredict

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AIM:

To analyze and predict the performance of CRISPR–Cas9 guide RNAs (gRNAs) targeting the LPL gene using the CRISPRredict tool, and to identify the most suitable gRNA with high on-target activity and low off-target risk.

INTRODUCTION:

CRISPR–Cas9 is an advanced genome-editing system that uses a guide RNA (gRNA) to direct the Cas9 enzyme to a specific DNA sequence. The accuracy and success of genome editing depend greatly on the efficiency and specificity of the designed gRNA. Inefficient or poorly designed gRNAs may lead to reduced editing performance or unintended modifications at off-target sites. CRISPRredict is a bioinformatics tool developed to estimate the activity and effectiveness of CRISPR–Cas9 gRNAs. It evaluates several sequence-based features such as GC percentage, secondary structure formation, genomic environment, and possible off-target interactions. In this study, CRISPRredict was applied to analyze potential gRNAs targeting the LPL (Lipoprotein Lipase) gene to determine the most efficient and reliable candidate for genome editing applications.

METHODOLOGY:

1. The LPL gene was chosen as the target for CRISPR–Cas9-mediated editing.
2. The genomic sequence of the LPL gene was uploaded into the CRISPRedict platform.
3. The tool generated multiple possible gRNA candidates targeting different sites within the gene.
4. Each candidate gRNA was assessed based on:
 - ✓ Predicted activity score
 - ✓ GC content
 - ✓ Target strand orientation
 - ✓ Self-complementarity potential
 - ✓ Predicted off-target interactions
5. The generated gRNAs were ranked according to their predicted efficiency scores.
6. The top-performing guide RNA was selected for detailed evaluation and interpretation.

QUESTION:

To predict the efficiency of CRISPR–Cas9 guide RNA using the CRISPRpredict tool.

RESULTS:



Fig 1: CRISPRpredict output showing ranked candidate gRNAs targeting the LPL gene with efficiency scores and sequence features.

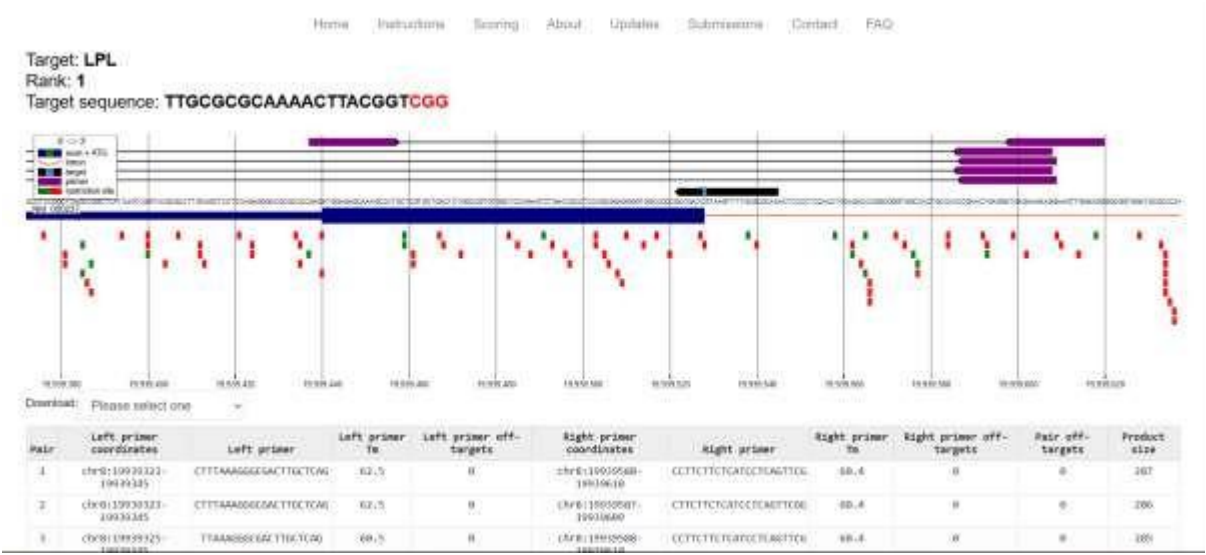


Fig 1.2: Genomic visualization of the top-ranked LPL gRNA with target sequence alignment and primer design details.

INTERPRETATION:

The CRISPRpredict analysis provided several potential gRNA sequences targeting various regions of the LPL gene, each showing different predicted activity levels. The highest-scoring gRNA demonstrated strong predicted efficiency, balanced GC content, and low self-complementarity, suggesting stable binding and effective Cas9 cleavage activity.

The selected gRNA:

- ✓ Specifically targets a defined region of the LPL gene
- ✓ Exhibits minimal predicted off-target binding
- ✓ Displays favorable sequence and thermodynamic characteristics

Genome visualization confirmed that the chosen guide RNA accurately aligns within the intended gene region without overlapping repetitive or undesirable genomic elements. Overall, the findings indicate that the selected gRNA is a strong candidate for precise and efficient CRISPR–Cas9 genome editing of the LPL gene.